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# Securing Certificate Management System Using Hyperledger Based Private Blockchain

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***Abstract*—Certificate digitalization has become a core part nowadays. It includes information about certificates where data is stored digitally. So far, most of the certificates are not digital which are difficult to maintain, and not highly secured. For these reasons, forgery of certificates and duplication of certificates are continuously happening. Blockchain gives us the opportunity of storing data that are highly secured and more efficient to use. Every transaction in this system is immutable which means it is quite impossible to change the data or hack the system. In this paper, a Hyperledger based low-cost and private blockchain is used for accomplishing the certificate system. This system restricts the access of users in some way but the owner is allowed to access all the functions of managing certificates. This privileges more security, privacy and authentication. Transactions have been made easier to be updated shortly and to insert the data into the blockchain. Deploying this blockchain-based certificate system will be beneficial for the digital world as it will resolve the issues of maintenance and indemnity.**

***Keywords—Certificate, Hyperledger, Blockchain, Chaincode***

## I. INTRODUCTION

Certificate is a valuable phenomenon in our life. The implications of certificates cannot be denied. Certificate enriches opportunity of employment, accessing community involvement benefits and better opportunities for retaining a current job. The deficiency of maintaining a secure system can cause fabrication of certificates. Certificate represents one's ability and by observing it authority can justify about the person. In the case of a fabricated certificate, the justification will also be inaccurate. For this we need one kind of system where fabrication of certificates can be repelled. Blockchain has introduced us to build this kind of system where the management of certificates will be much more feasible and secured. Management of certificates mainly focuses on digitalization of certificates. When we alter the information of certificates into digital data, it is known as digitalization of certificates. Through digitalization we can prevent manipulated certificates, the illegal access of other's certificates. The drawbacks of public blockchain are that anyone can see information which is not required and we cannot control the access of users. The speed of public blockchain is relatively low. On the other hand, Ethereum requires transaction fees. Though Ethereum is the second most decentralized cryptocurrency in the world after Bitcoin, it has some issues. The issues include high traffic in the network, not fully immutable and some well-known problems in the programming language of Ethereum. In this paper, we used Hyperledger blockchain which is an open source and private blockchain and everyone cannot access every part. It has permissioned mode of operation and is used to build enterprise blockchains.

Hyperledger not only solves trust issues in the network but also ensures a direct approach through peer-to-peer transmission and decentralization. It's high-level security, interoperable nature, modular design and high-end API (Application Programming Interface) support and cryptocurrency-agnostic structure makes it useful across different industries.

## II. LITERATURE REVIEW

Blockchain is a relatively new technology that is now being used in a variety of fields. In paper [1] the researchers proposed a hypothetical cloud and blockchain-based certificate verification system that could provide a potential solution for academic certificate issuing and verification in areas where cryptocurrencies are prohibited. In their paper, they didn't propose architecture for the graduate in the various stages such as undergraduate, postgraduate and others. And also, evaluation of performances of their architecture isn't presented in numerous parameters. In paper [2] other authors introduced the educational certificate blockchain system that can facilitate low latency and high throughput while also providing a means for query optimization. Though they used digital infrastructure to handle educational certificates, this theory can be applied to numerous fields such as proof of identity, proof of professional qualifications etc. For COVID-19 test takers, the researchers in paper [4] includes digital medical passports (DMP) and immunity certificates which demonstrate smart contracts based on the Ethereum blockchain that have been basically developed and tested to retain a digital medical identity, allowing for a quick and trusted response from the adequate medical authorities. They utilized an Ethereum-based blockchain, which is risky and can lead to vulnerabilities, breakdowns, hacks, and other issues. The authors of paper [7] construct an examination of blockchain adoption in relation to the process of creating meaningful certificates for higher education students. In their proposed DASC (dApp for Smart Certificates), they didn't show the evaluation and validation of uses of their desired solution. In addition, the application of this project in the higher education sector is also missing. The Paper [9] is about to provide a server to store job training documents and certificates based on the blockchain platform open to the public. Because IPFS (Inter Planetary File System) is constantly growing and adopted by a lot of users, the storage issue is not solved. Furthermore,

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the trust issue also remains here though data is stored in a distributed and transparent environment of blockchain. In paper [10], the authors presented a certificate management platform for performance evaluation during recruitment using Bitcoin support and Ethereum. There is systemic risk in Bitcoin, as well as a lack of warning along with inadequate functions. Ethereum, on the other hand, is a public blockchain feature that enables anybody to view the data contained within the blocks. More structured blockchain-based authentication and secure certificate revocation scheme is submitted by the authors of paper [11] which mainly constructed on the signature module of X.509 digital certificate by replacing certificate hash value and a consensus algorithm is given for better efficiency. On the proposed algorithm, they need to check it with more network scenarios and implementation of the scheme in blockchain network based is also missing. Applying smart contracts for development to better the efficiency is also needed. In paper [12], the Blockchain for education platform is presented as a realistic method for issuing, verifying, and exchanging certificates. The system has monetary overhead since it runs on the Ethereum network. The whole system is considered insecure as Ethereum is a public and permissionless blockchain and therefore, anyone can access the certificates. The researchers of paper [13] demonstrates the blockchain for education platform as a realistic option for issuing, supervising, authenticating, and exchanging certificates in this showcase. The blockchain for education platform is built on the Ethereum blockchain and supports the certification process through smart contracts. This system also suffers from the same aforementioned difficulties of [12]. Also, this system utilizes IPFS to store certificates, which increase its vulnerability as anyone who has the Content Identifier (CID) of the certificate uploaded to the IPFS can download the certificate and fetch confidential information from it. The authors of [14] suggests a blockchain-based solution for data unity and openness, data access efficiency, long-term records preservation, and, most importantly, a system that is less affordable and faster. There are drawbacks to not prioritizing the significance of smart contract security, as well as an analysis of its vulnerabilities and proper PKI management.

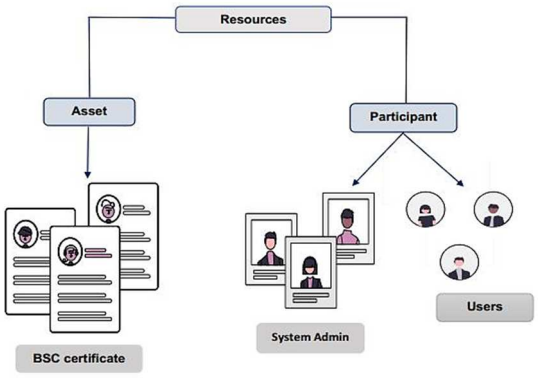
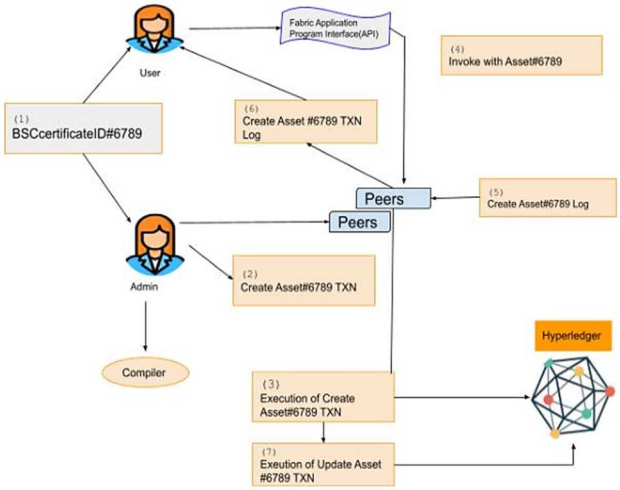


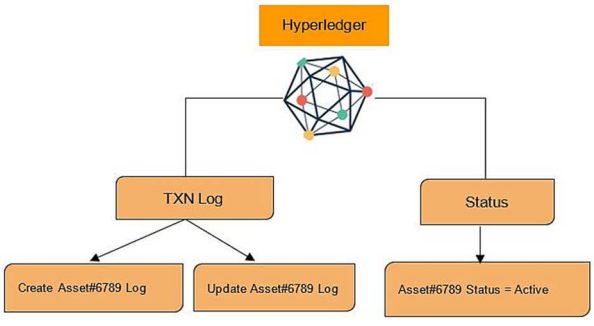

Fig. 1: Classified System Resources.

## III. Proposed Approach

In this paper, we implemented a certificate digitalization system using a private Hyperledger blockchain where every transaction is immutable and cannot be accessed by everyone. In Fig. 1 we illustrate the system resources of our Hyperledger based system where system admin and end user roles are defined as participants and by getting required information an assetID and an asset is created. Admin or owner has the ability to create a certificate asset and users have some restrictions while using the system. Every certificate asset has an asset ID by which that can be uniquely identified. So those certificates which have an asset ID are considered as a real one. Necessary information will be collected from the user. If there is some mistake or errors in the required information, it can be reformed. All transactions will be stored which cannot be changed or modified. That means, if some information gets some errors and if the information is corrected later; both the error and the updated transactions will be stored. There is also a transaction which is called 'view transactions' where if anybody sees the certificate, the information will be stored in transactions. For this, illegal access of certificates can be easily determined and thus the security is very tight here. Unlike the other confidential documents such as land registry, ensuring the security of educational certificates is challenging as there are different tiers in the hierarchy of educational institutions. Therefore, constructing a unified blockchain system for storing certificates is quite difficult.




(a)




(b)

Fig. 2: System Architecture of Our Proposed Hyperledger Based Certificate System: (a) System's Information Flow, (b) Interaction with Blockchain.

As Hyperledger is a consortium blockchain, our proposed business network can be partitioned into corresponding tiers relevant to the type of the certificate.

## IV. METHODOLOGY

In this system Hyperledger based private blockchain is accomplished. Every asset stored in the blockchain will create a hash which is unique in nature. If anybody changes the hash it will not be matched with the real one. So, it's very much difficult to modify anything. The create and updating information will be recorded as a transaction. The entire structure of the system is illustrated in Fig. 2 which is mentioned below:

### *A. Creating the Assets:*

Every certificate will be stored as an asset in the Hyperledger blockchain. The asset contains the valuable information of certificates such as name, year, CGPA and so on. These assets are collected and accumulated in the asset registry of the private blockchain. It is identified by BSc certificate ID in our system which indicates the particular certificate's circumstantial information and by this id number, it is indexed in the asset registry. Asset registry is nothing but the necessary information collected from the users and stored in the organization asset for further process. However, creating and modifying asset information can only be done by the asset transaction which proves the unchangeable features of blockchain. In this way, history can be preserved.

### *B. Execution of Transactions:*

Every sustainable operation of our proposed blockchain-based protocol is performed, just like every other blockchain system. To create, access and update the assets this system needs to execute transactions are being used. Fig. 3 demonstrate the execution of transactions in this system. Once a transaction is placed in one of the chain's blocks, it becomes immutable, which ensures it can't be changed without a massive attack like the majority attack, which is almost impossible to conduct. Whenever an asset is created that means a new certificate is added to the blockchain it will be stored as a transaction. If someone gains access to the certificate, the report will be saved as well. The information transaction will be performed out if it has to be updated. As a result, no one will be able to change the certifications of other people. When a user performs a transaction, the ID is compared to the asset ID. If it isn't matched, it will be taken into account as an illegal or unethical attempt.

### *C. Determining Participant Types:*

In order to determine the system's access control, participants are expected in our system. System admin and end user - are the two types of participants to manage the system. The next subsections go over all of the participants' roles.

**Transaction Operation for Certificate Asset Creation, Update, and Emitting event**

Implementation of transaction for creating a BSC certificate
Input: BSCcertificateId, registrationNo, session, firstName, lastName, age, idNo, fathersName, mothersName, cgpa, department, universityName, country, date

```
BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate ← Entity  of factory.newResource(NS,'BSCcertificate',BSCdata.BSCcertificateId)

BSCcertificate.owner ← BSCdata.owner
BSCcertificate.registrationNo ← BSCdata.registrationNo
BSCcertificate.session ← BSCdata.session
BSCcertificate.firstName ← BSCdata.firstName
BSCcertificate.lastName ← BSCdata.lastName
BSCcertificate.age ← BSCdata.age
BSCcertificate.idNo ← BSCdata.idNo
BSCcertificate.fathersName ← BSCdata.fathersName
BSCcertificate.mothersName ← BSCdata.mothersName
BSCcertificate.cgpa ← BSCdata.cgpa
BSCcertificate.department ← BSCdata.department
BSCcertificate.universityName ← BSCdata.universityName
BSCcertificate.country ← BSCdata.country
BSCcertificate.date ← BSCdata.date

event ← Entity of factory.newEvent(NS,'BSCcertificateCreated')
    event.BSCcertificateId ← BSCdata.BSCcertificateId
    emit(event)
 return BSCcertificateRegistry.add(BSCcertificate)
```

Implementation of transaction for updating a BSC certificate
Input: BSCcertificateId, newsession, newfirstName, newlastName, newfathersName, newmothersName, newcgpa

```
BSCcertificate Registry ← Entity  of getAssetRegistry('org.BSC.assetTxn.BSCcertificate')
factory ← Instance of  getFactory()
NS ← Instance of org.BSC.assetTxn

BSCcertificate.BSCcertificateId ← updatesData.BSCcertificateId
BSCcertificate.session ← updatesData.newsession
BSCcertificate.firstName ← updatesData.newfirstName
BSCcertificate.lastName ← updatesData.newlastName
BSCcertificate.fathersName ← updatesData.newfathersName
BSCcertificate.mothersName ← updatesData.newmothersName
BSCcertificate.cgpa ← updatesData.newcgpa

 event ←  factory.newEvent(NS,'BSCcertificateUpdated')
    event.BSCcertificateId ← updatesData.BSCcertificateId
    emit(event)
  return bC.update(BSCcertificate)
```

Fig. 3: Transactional Operation for Implementing Creating and Updating Certificate.

1. System Admin: The system administrator maintains overall control and has access to all of the system's data. Admin has access to the assets and historian records. He can also make adjustments, such as creating it or updating it. Admins have the potential to customize new participants as well as inspect the record. To prevent any form of breach, the system administrator can take appropriate measures.

2. End Users: End users have various restrictions on how they can access the system as a whole. They can only retrieve data related to their certificates but cannot modify it. The ability of end users to create assets is likewise limited.

### *D. Maintaining Assets in the Asset Registry:*

All of our system's assets information will be retained in the asset registry which is depicted in Fig. 4. The asset registry will be adjusted whenever the admin introduces a new asset. Because blockchain has immutable properties, no asset can be truly erased. If a system administrator or end user attempts to access important data about a certain asset, they must input the asset ID into this registry. However, if it is attempted to be deleted, the status will be changed to "deleted". But the asset will remain in place. Furthermore, the updated information will also be stored in the asset registry.




Fig. 4: Asset Registry for the Proposed Certificate System.

### *E. Fetching Transactions from the Historian Records:*

In our system, there are primarily two transactions: create transaction and update transaction. Transactions are recorded in blocks in the blockchain system, which are unchangeable. For this feature, the transaction's information can be acquired from the historian record. Historian records are indexed by a particular ID. So, in order to retrieve the information, the transaction ID and historian ID is needed. It may be determined whether a transaction exists or has already been confirmed by a user. It may also be verified if there has been an update by using this functionality inside the asset registry and historical record. We can classify the illegal behaviors based on the transaction records. Because each user's certificate id is unique, every attempt to view or retrieve information from another user will result in a transaction which will be recognized in the historian record.

### *F. Querying Transactions in Historian Records:*

All assets are maintained in the asset registry, and all transactions are preserved in the historian record in our system. Aside from transaction ID, the historical record is additionally indexed by the participant ID accountable for the transaction's execution and the transaction's type. Whenever we need to access asset information from the asset registry, we use a query. The authentication of Certificate validation by transaction query is illustrated in Fig. 5. So, a query is also needed to retrieve transactional information from the historian record.

### *G. Establishing Privacy Settings and Regulations:*

To access all of the assets, we needed distinct permissions. Here, we require both permissions to access the asset register and to retrieve information. Whereas, End users do not have permissions to all rights, but administrators do. So, administration has a major effect in establishing privacy setting regulations. Participant instances are kept in the admin composer for securing the certificate system. By this admin can authenticate users. User can read and update his/her information by historian record but user can only read the information of asset registry. Setting permissions on the historian record allows us to control user access which is shown in Fig. 6. The system does not grant authorization to an uninvited user, i.e., a person who does not have access, as it utilizes the access control feature of

**Validating Certificate authentication by access transaction and access registers**

```
Query AccessHistorianRecord:
        description ← "Returns Historian Records"
              statement ← SELECT org.hyperledger.composer.system.
                          HistorianRecord where (transactionType == _$input)
Query AccessAssetRegistry:
              description ← "Returns Asset Registry Elements
              statement ← SELECT org.hyperledger.composer.system.AssetRegistry
                          where(assetID ==$input)
acessSpecificAsset:
                    bnUtil ← Business Network Definition of Certificate System
              assReg ← bnUtil.connection.query('AccessAssetRegistry')
              qry ← bnUtil.connection.buildQuery(assReg)
              return bnUtil.connection.query(qry, {input: assReg#ID)
acessUpdateAssetTransaction:
                    bnUtil ← Business Network Definition of Certificate System
              histRecs ← bnUtil.connection.query('AccessHistorianRecord')
              qry← bnUtil.connection.buildQuery(assReg)
              return bnUtil.connection.query(qry, {input: 'updateCertificateInfo')
checkAssetValidity:
              Input: Asset/BSCcertificateID
              asset ← accessSpecificAsset(assReg#ID)
              if asset IS NOT void
                then historianRecs ← accessUpdateAssetTransaction()
                    for record in historianRecs:
                        if record.BSCcertificateID IS assReg#ID
                        return record
              else
                  return VOID
```

Fig. 5: Authentication of Certificate Validity by Transaction Query.

Hyperledger. However, if someone tries to access by asset ID, it can be prevented by the rules defined using the CRUD (create, read, update, delete) operations in the access control module. We can also establish permissions for fetching each transaction, such as create and update.

**Participant Creation, Assigning Identities and Providing Participants the Access to Execute Transactions**

```
Assigning Identities to End-Use Participant:
    participantInstance←composer.participant.add(class:EndUser,participantkey:6789,fname: Shamma,
lname: Dilshad, contactNo: 01718421092, email: shammadilshad@gmail.com)
    composer.identity.issue(participantInstance,admin@certificate-securing-ledger)

Access Control for System Admin and End Users:
Rule  AdminSystemResource:
      description← "Grant business network administrators full access to system resources"
      participant← "org.hyperledger.composer.system.NetworkAdmin"
      operation← ALL
      resource← "org.hyperledger.composer.system.**"
      action← ALLOW

Rule EndUserPermissionHistorianRecordREAD_CREATE:
      description← "Can VIEW and CREATE HistorianRecord inside the Historian"
      participant← "org. certificate-securing-ledger.participants.EndUser"
      operation← READ, CREATE
      resource← "org.hyperledger.composer.system.HistorianRecord"
      action← ALLOW

Rule EndUserPermissionAssetRegistryREAD:
      description← "Can only VIEW Assets inside AssetRegistry"
      participant← "org. certificate-securing-ledger.participants.EndUser"
      operation← READ
      resource← "org.hyperledger.composer.system.AssetRegistry"
      action← ALLOW


Rule EndUserPermissionUpdateCertificateInfoCREATE:
      description←"Can EXECUTE UpdateCertificiateInfo transaction"
      participant← "org. certificate-securing-ledger.participants.EndUser"
      operation← READ
      resource← "org. certificate-securing-ledger.certificateInfo. UpdateCertificateInfo"
      action← ALLOW
```

Fig. 6: Access Control Limiting Blockchain Access.

## V. RESULT ANALYSIS

We gathered relevant student information from Bangladesh Army International University of Science and Technology (BAIUST) in order to create certificates for our system's inspection. In the following Table 1, a sample of student information is given.

TABLE I: Arbitrary Student Information for System Inspection Purpose

| Serial No | ID | Student Name | CGPA | Dept. | Institution |
|---|---|---|---|---|---|
| 01 | 1106028 | Shamma Dilshad | 3.85 | CSE | BAIUST |
| 02 | 1106026 | Saima Ahmed | 3.78 | CSE | BAIUST |
| 03 | 1205016 | Ehsanul Haque | 3.90 | EEE | BAIUST |
| 04 | 1205297 | Zerin Tasnim | 3.80 | CSE | BAIUST |
| 05 | 1306013 | Fatema Simi | 3.70 | BBA | BAIUST |
| 06 | 1306016 | Syed Resalat | 3.65 | BBA | BAIUST |

Admin created assets using the data from Table I and then secured the assets in the blockchain as a historian record shown in Fig. 7.

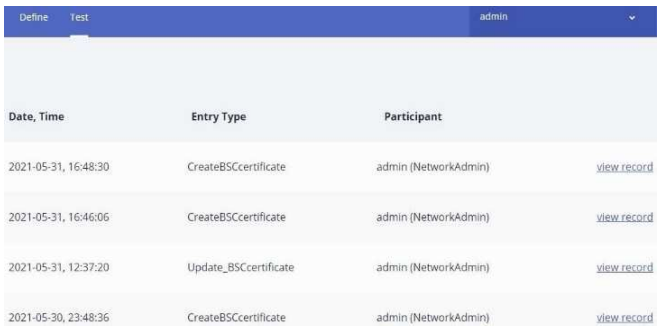



Fig. 7: List of Successful Transactions in Historian Record.

Assets are created using the CreateBSCcertificate transaction, and they can also be updated using the Update_BSCcertificate transaction, as shown in the illustration of Fig. 8 and Fig. 9.

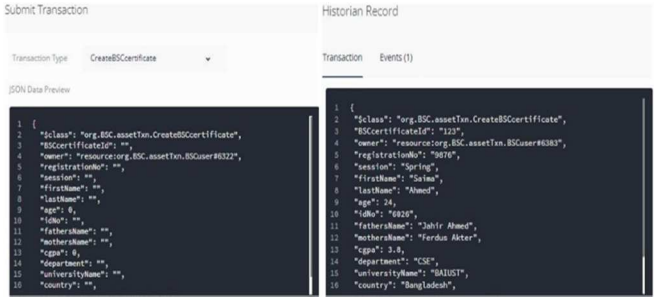



Fig. 8: Submission of create_asset_transaction and Accessing the Same Transaction Details from Historian Record.

We also implemented our system on Ethereum and Bitcoin to analyze the computational delay. The remix IDE and the solidity programming language are used for Ethereum. We measured the computational latency by implementing our system in Python using the 'python-bitcoinlib' library [15]. The computational latency is measured by the 'block.number' and 'block.timestamp' methods in Ethereum

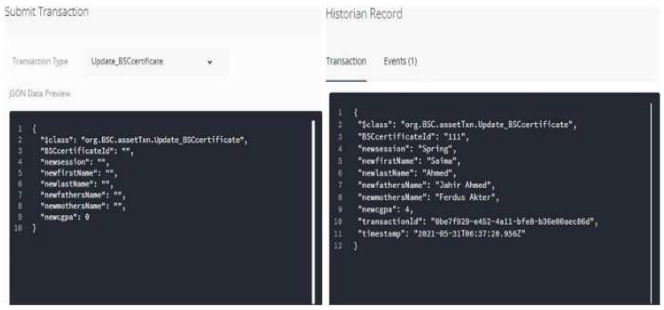



Fig. 9: Submission of update_asset_transaction and Accessing the Same Transaction Details from Historian Record.

whereas in case of Bitcoin, 'time' and 'now' functions of the Python library 'time' is used.

For estimating the effectiveness of Hyperledger in our certificate system, we gathered information from the BAIUST database and checked the differences of required time inserting the information of Bitcoin based blockchain and Ethereum framework. Fig. 10 gives the comparison of computation time for three blockchain systems which is based on the data of Table II. After checking all the things, we came to the conclusion that Hyperledger is best for this system.

TABLE II: Time Needed to Execute Transactions in Various Blockchain Technologies

| Number of Transactions | Time for Hyperledger (in second) | Time for Ethereum (in second) | Time for Bitcoin (in second) |
|---|---|---|---|
| 350 | 7.53 s | 40.9 s | 42.89 s |
| 300 | 6.43 s | 32.89 s | 35.675s |
| 250 | 4.58 s | 25.5 s | 27.68 s |
| 200 | 3.26 s | 19.85 s | 21.54 s |
| 150 | 2.45 s | 12.9 s | 15.26 s |
| 100 | 1.98 s | 5.35 s | 7.185 s |

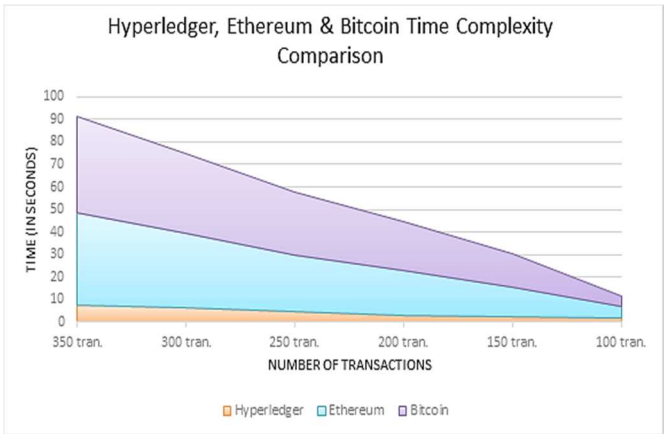



Fig. 10: Comparison of Computation Time for Three Blockchain Systems.

## VI. CONCLUSION

Our system is more stable and secure by using Hyperledger Blockchain. Hyperledger's permissioned and private nature aids the system in executing its operations more reliably. Access to other people's certificates is restricted in the certificate system, which is not feasible in any other permissionless blockchain. A certificate which is used to validate the system user's authenticity is automatically stored on a user's device when he or she is added to the Hyperledger system. As the validating certificates are device-specific, no one can duplicate them onto another device. As a consequence, we can prevent our system from being hacked through phishing or spoofing. Ethereum is not suited for building a certificate system since it requires cryptocurrency to execute any form of transaction and transaction data is generally accessible. In addition, the amount of storage available is minimal. Moreover, the Hyperledger-based authorized network makes use of the decentralized network's historical record, which ensures that a recorded transaction cannot be changed in any way.

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